

# Maze Solver Software Development Schedule (10 Weeks)

## Project Information

**Project Name:** Maze Solver Software

**Programming Language:** Java

**Course:** Software Development 1

**ID:** 11240321733

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## Week 1: Project Planning and Requirement Analysis

### What I Will Do

- Analyze project requirements.
- Study maze structures and solving concepts.
- Identify software features and modules.
- Prepare project objectives and scope.
- Create development roadmap.

### What I Need to Learn

- SDLC Basics
- Maze Fundamentals
- Requirement Analysis
- Git and GitHub Basics

### GitHub Upload

- Project Proposal
- Requirement Analysis Document
- Initial README File

### What I Will Demonstrate

- Project Idea
  - Software Features
  - Development Plan
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## **Week 2: Software Design**

### **What I Will Do**

- Design software architecture.
- Create class diagram.
- Design workflow.
- Plan user interface.

### **What I Need to Learn**

- Object-Oriented Programming (OOP)
- UML Class Diagram
- Java Classes and Objects
- Software Design Principles

### **GitHub Upload**

- System Design Document
- Class Diagram
- Project Structure

### **What I Will Demonstrate**

- Software Architecture
  - Workflow Design
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## **Week 3: User Interface Development**

### **What I Will Do**

- Create application window.
- Develop Main Menu.
- Add Start, Help and Exit options.
- Design navigation system.

### **What I Need to Learn**

- Java Swing
- JFrame
- JButton
- Event Handling

## **GitHub Upload**

- Main Menu Source Code
- GUI Design Files

## **What I Will Demonstrate**

- Working Main Menu
  - Navigation System
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# **Week 4: Maze Board Development**

## **What I Will Do**

- Create maze grid.
- Design walls and paths.
- Display maze visually.
- Improve interface appearance.

## **What I Need to Learn**

- Java Graphics
- JPanel
- Drawing Components
- Layout Management

## **GitHub Upload**

- Maze Board Module
- Grid Visualization Module

## **What I Will Demonstrate**

- Maze Visualization
  - Software Interface
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## **Week 5: Player Movement Implementation**

### **What I Will Do**

- Implement player movement.
- Add keyboard controls.
- Update player position.
- Track movement inside maze.

### **What I Need to Learn**

- Key Events
- Event Handling
- Coordinate System
- Game Logic Basics

### **GitHub Upload**

- Player Movement Module
- Control System

### **What I Will Demonstrate**

- User Navigation in Maze
  - Basic Working Software
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## **Week 6: Maze Validation System**

### **What I Will Do**

- Prevent movement through walls.
- Validate player actions.
- Display warning messages.

### **What I Need to Learn**

- Conditional Logic
- Error Handling
- Validation Techniques

## **GitHub Upload**

- Validation Module
- Error Handling Module

## **What I Will Demonstrate**

- Wall Collision Detection
  - Rule Validation System
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# **Week 7: Software Features Enhancement**

## **What I Will Do**

- Add Move Counter.
- Add Game Timer.
- Add Restart Option.
- Improve usability.

## **What I Need to Learn**

- Java Timer
- Labels and Dynamic Updates
- State Management

## **GitHub Upload**

- Timer Feature
- Move Counter Feature
- Restart Function

## **What I Will Demonstrate**

- Timer Working
  - Move Counter
  - Restart Functionality
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## **Week 8: Auto Solver Development**

### **What I Will Do**

- Implement maze solving algorithm.
- Create Auto Solve button.
- Show solution path step-by-step.

### **What I Need to Learn**

- DFS Algorithm
- BFS Algorithm
- Pathfinding Concepts

### **GitHub Upload**

- Solver Algorithm Module
- Auto Solver Feature

### **What I Will Demonstrate**

- Automatic Maze Solving
  - Path Visualization
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## **Week 9: Software Finalization**

### **What I Will Do**

- Add Goal Detection.
- Add Success Message.
- Improve UI Design.
- Add About Section.

### **What I Need to Learn**

- Software Testing Basics
- UI Improvement Techniques
- User Experience Design

## **GitHub Upload**

- Goal Detection Module
- Updated User Interface
- About Section

## **What I Will Demonstrate**

- Complete Software Workflow
  - Winning Screen
  - Improved Interface
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# **Week 10: Testing and Final Release**

## **What I Will Do**

- Perform complete testing.
- Fix bugs and errors.
- Prepare documentation.
- Prepare final presentation.

## **What I Need to Learn**

- Software Testing
- Debugging
- Documentation Writing
- Presentation Skills

## **GitHub Upload**

- Final Source Code
- Documentation
- User Manual
- Screenshots
- Version 1.0 Release

## **What I Will Demonstrate**

- Complete Software Application
- All Features Working
- Final Project Presentation